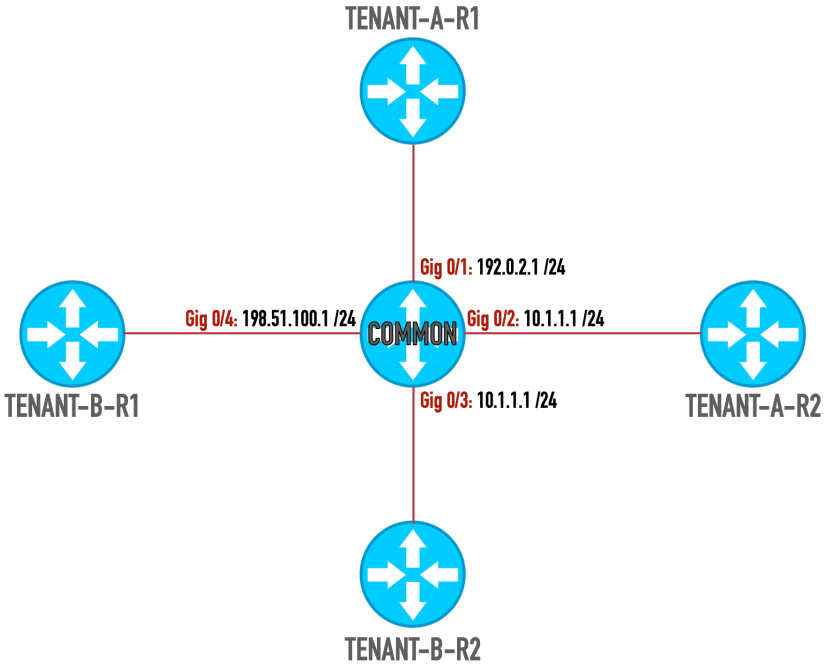
Virtual Routing & Forwarding (VRF)

Topology



Initial Configuration Commands

COMMON:

```
enable
conf t
host COMMON
no banner motd
no banner login
no banner exec
no banner incoming
line vty 0 15
password cisco
login
exec-timeout 0 0
line con 0
logging synchronous
no boot network
exit
no ip-domain lookup
end
copy run star
```

Copyright 2024, Kevin Wallace Training

(##NOTE: Even though the topology shows five routers, we only need to configure the "COMMON" router.)

Lab Tasks

1. Create a multi-protocol instance for each tenant that we have, as well as specify which address family they are a part of on the COMMON router.
2. Assign each of our interfaces, on the COMMON router, to one of these VRF instances based upon our topology.
3. Let's see what VRFs were created, and what interfaces apply to them.
4. Let's take a look at the routing tables for both TENANT-A and TENANT-B.
5. In addition to the routing tables that we have, one for TENANT-A and one for TENANT-B, there is also a global routing table. We can see this by our traditional "show ip route" command. As we can see, there are currently no global routes. Let's create a loopback interface with the ipv4 address of 5.5.5.5 /32.
6. Let's see if the loopback address shows up in our global routing table. Let's also see if the loopback address shows up in the routing tables for TENANT-A and TENANT-B.

Solution

Step 1: Create a multi-protocol instance for each tenant that we have, as well as specify which address family they are a part of on the COMMON router.

```
COMMON>en
COMMON#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
COMMON(config)#vrf definition TENANT-A
COMMON(config-vrf)#address-family ipv4
COMMON(config-vrf-af)#exit
COMMON(config-vrf)#vrf definition TENANT-B
COMMON(config-vrf)#address-family ipv4
COMMON(config-vrf-af)#exit
COMMON(config-vrf)#exit
```

Copyright 2024, Kevin Wallace Training

Step 2: Assign each of our interfaces, on the COMMON router, to one of these VRF instances based upon our topology.

```
COMMON(config)#int gig 0/1
COMMON(config-if)#vrf forwarding TENANT-A
COMMON(config-if)#ip address 192.0.2.1 255.255.255.0
COMMON(config-if)#no shutdown
COMMON(config-if)#int gig 0/2
COMMON(config-if)#vrf forwarding TENANT-A
COMMON(config-if)#ip address 10.1.1.1 255.255.255.0
COMMON(config-if)#no shutdown
COMMON(config-if)#int gig 0/3
COMMON(config-if)#vrf forwarding TENANT-B
COMMON(config-if)#ip address 10.1.1.1 255.255.255.0
COMMON(config-if)#no shutdown
COMMON(config-if)#int gig 0/4
COMMON(config-if)#vrf forwarding TENANT-B
COMMON(config-if)#ip address 198.51.100.1 255.255.255.0
COMMON(config-if)#no shutdown
COMMON(config-if)#end
```

(##Notice what would typically be an ip address overlap with int gig 0/2, is not in this case because int gig 0/2 and int gig 0/3 are in different virtual router instances.)

Step 3: Let's see what VRFs were created, and what interfaces apply to them.

```
COMMON#show ip vrf
  Name                              Default RD            Interfaces
  TENANT-A                          <not set>             Gi0/1
                                                          Gi0/2
  TENANT-B                          <not set>             Gi0/3
                                                          Gi0/4
```

Step 4: Let's take a look at the routing tables for both TENANT-A and TENANT-B.

```
COMMON#show ip route vrf TENANT-A

Routing Table: TENANT-A
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static
route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
```

Copyright 2024, Kevin Wallace Training

```
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

      10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C        10.1.1.0/24 is directly connected, GigabitEthernet0/2
L        10.1.1.1/32 is directly connected, GigabitEthernet0/2
      192.0.2.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.0.2.0/24 is directly connected, GigabitEthernet0/1
L        192.0.2.1/32 is directly connected, GigabitEthernet0/1
COMMON#show ip route vrf TENANT-B

Routing Table: TENANT-B
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static
route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

      10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C        10.1.1.0/24 is directly connected, GigabitEthernet0/3
L        10.1.1.1/32 is directly connected, GigabitEthernet0/3
      198.51.100.0/24 is variably subnetted, 2 subnets, 2 masks
C        198.51.100.0/24 is directly connected, GigabitEthernet0/4
L        198.51.100.1/32 is directly connected, GigabitEthernet0/4
```

(##We are only seeing ip addresses that are assigned to TENANT-A interfaces because it is isolated from TENANT-B.)
(##We are only seeing ip addresses that are assigned to TENANT-B interfaces because it is isolated from TENANT-A.)

Step 5: In addition to the routing tables that we have, one for TENANT-A and one for TENANT-B, there is also a global routing table. We can see this by our traditional "show ip route" command. As we can see, there are currently no global routes. Let's create a loopback interface with the ipv4 address of 5.5.5.5 /32.

```
COMMON#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
```

Copyright 2024, Kevin Wallace Training

```
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static
route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

COMMON#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
COMMON(config)#int lo0
COMMON(config-if)#ip address 5.5.5.5 255.255.255.255
COMMON(config-if)#end
```

(##Right now we don't have any global routes, until we create the loopback interface of 5.5.5.5.)

Step 6: Let's see if the loopback address shows up in our global routing table. Let's also see if the loopback address shows up in the routing tables for TENANT-A and TENANT-B.

```
COMMON#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static
route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

      5.0.0.0/32 is subnetted, 1 subnets
C        5.5.5.5 is directly connected, Loopback0
COMMON#show ip route vrf TENANT-A

Routing Table: TENANT-A
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static
route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set
```

Copyright 2024, Kevin Wallace Training

```
      10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C        10.1.1.0/24 is directly connected, GigabitEthernet0/2
L        10.1.1.1/32 is directly connected, GigabitEthernet0/2
      192.0.2.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.0.2.0/24 is directly connected, GigabitEthernet0/1
L        192.0.2.1/32 is directly connected, GigabitEthernet0/1
COMMON#show ip route vrf TENANT-B

Routing Table: TENANT-B
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static
route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

      10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C        10.1.1.0/24 is directly connected, GigabitEthernet0/3
L        10.1.1.1/32 is directly connected, GigabitEthernet0/3
      198.51.100.0/24 is variably subnetted, 2 subnets, 2 masks
C        198.51.100.0/24 is directly connected, GigabitEthernet0/4
L        198.51.100.1/32 is directly connected, GigabitEthernet0/4
```

(##As of right now we have three routing tables, and they are all isolated from each other.)
(##This is the global routing table.)
(##This is the routing table for TENANT-A.)
(##This is the routing table for TENANT-B.)

Copyright 2024, Kevin Wallace Training